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How Sampling Impacts the Robustness of Stochastic Neural Networks

Sina Däubener and Asja Fischer
Department of Computer Science
Ruhr University Bochum, Germany
{sina.daeubener, asja.fischer}@rub.de

Abstract

Stochastic neural networks (SNNs) are random functions whose predictions are gained by averaging over multiple realizations. Consequently, a gradient-based adversarial example is calculated based on one set of samples and its classification on another set. In this paper, we derive a sufficient condition for such a stochastic prediction to be robust against a given sample-based attack. This allows us to identify the factors that lead to an increased robustness of SNNs and gives theoretical explanations for: (i) the well known observation, that increasing the amount of samples drawn for the estimation of adversarial examples increases the attack's strength, (ii) why increasing the number of samples during an attack can not fully reduce the effect of stochasticity, (iii) why the sample size during inference does not influence the robustness, and (iv) why a higher gradient variance and a shorter expected value of the gradient relates to a higher robustness. Our theoretical findings give a unified view on the mechanisms underlying previously proposed approaches for increasing attack strengths or model robustness and are verified by an extensive empirical analysis.

1 Introduction

Since the discovery of adversarial examples [Biggio et al., 2013, Szegedy et al., 2014], a significant amount of research was dedicated to hinder attacks [e.g. Madry et al., 2018, Papernot et al., 2016, Zhang et al., 2019], to enhance attack strategies [e.g. Athalye et al., 2018, Akhtar and Mian, 2018, Carlini and Wagner, 2017b, Uesato et al., 2018] or to derive ways to certify model robustness [e.g. Cohen et al., 2019, Lécuyer et al., 2019]. Robustness guarantees often specify an ϵ-ball around input points in which perturbations do not lead to a label change [Hein and Andriushchenko, 2017, Croce and Hein, 2020]. The maximal possible radius of such an ϵ-ball corresponds to the distance of the input point to the nearest decision boundary, which on the other hand is equal to the length of the smallest perturbation vector that leads to a misclassification (c.f. figure 1a)). Such a robustness analysis assumes that the decision boundaries are fixed and that the attacker is able to estimate (at least approximately) this minimal perturbation vector, which is a reasonable assumption for deterministic networks but usually does not hold for stochastic neural networks (SNNs).

Stochastic neural networks, and stochastic classifiers more generally, are random functions and predictions are given by the expected value of the random function for the given input. In practice, this expectation is usually not tractable and hence it is approximated by averaging over multiple realizations of the random function. This approximation leads to the challenging setting where predictions, decision boundaries, and gradients become random variables themselves. Hence, under an adversarial attack, the decision boundaries used for calculating the adversarial example and those used when predicting the label of the resulting adversarial example differ. This means that the attacker can not estimate the optimal perturbation direction i.e., the direction to the closest decision boundary of the network that will be sampled during inference c.f. figure 1 b), c).

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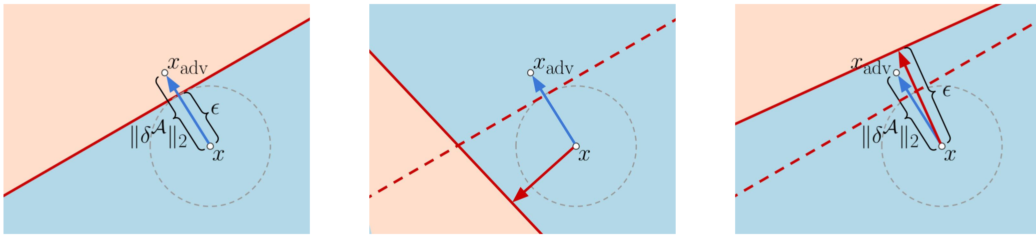

(a) (b) (c)

Figure 1: (Un-)successful attacks on a binary stochastic classifier with a linear decision boundary. a) an adversarial example $x_{\text{adv}} = x + \delta^{\mathcal{A}}$ is created by shifting x in the direction of the closest decision boundary, indicated by the blue arrow. ϵ is the minimal needed attack length and the attack can only be an adversarial example if $\|\delta^{\mathcal{A}}\|_2 \geq \epsilon$ holds. b) and c) show the stochastic decision boundaries during attack (dashed) and inference (solid). The red arrows indicate the shortest direction to the latter, respectively. In b) $\delta^{\mathcal{A}}$ moves x even further away from the decision boundary used during inference, while in c) the magnitude of $\delta^{\mathcal{A}}$ is too short to result in a successful attack.

In this paper, we study how robustness of SNNs arises from this misalignment of the attack direction and the optimal perturbation direction during inference that results from the stochasticity inherent to stochastic classifiers.

We make the following contributions: First, we derive a sufficient condition for a SNN prediction relying on one set of samples to be robust against an attack that was calculated on a second set of samples. Second, we discuss how model properties and sample sizes impact this condition which does not only allows us to answer the questions stated in the abstract but also to explain the success of recently proposed defense mechanism from a simple unifying geometric perspective. Lastly, we conduct an empirical analysis that demonstrates that the novel theoretical insights perfectly match what we observe in practice.

2 Related work

Several works proposed stochastic defense mechanism to increase adversarial robustness [e.g. Raff et al., 2019, Xie et al., 2018]. Athalye et al. [2018] linked their success to gradient obfuscating during the attack and showed that increasing the number of samples for approximating the gradient during attack leads to a sever decrease in adversarial accuracy.

However, SNNs were still found to have an increased robustness even w.r.t. stronger attacks [Yu et al., 2021, He et al., 2019, Eustratiadis et al., 2021, Jeddi et al., 2020]. Their success was attributed to different effects of stochasticity, e.g. model smoothing [Liu et al., 2018, Addepalli et al., 2021] or diversification of the gradients [Lee et al., 2022, Bender et al., 2020]. While a lot of work analyzed the robustness of deterministic neural networks [e.g. Madry et al., 2018, Croce et al., 2019, Croce and Hein, 2020, Dabouei et al., 2020, Yang et al., 2022], the robustness of SNNs is less well understood. One line of research focused on the robustness of Bayesian neural networks (BNNs). Wicker et al. [2020] certified robustness of BNNs using interval bound propagation techniques which they later also employed to derive guarantees for the robustness of BNNs with modified adversarial training [Wicker et al., 2021]. Moreover, Carbone et al. [2020] investigate robustness in the infinite-width infinite-sample limit. Lastly, Pinot et al. [2019] derived theoretical robustness guarantees for randomized networks, where the randomization is based on additive noise from an exponential family distribution. This leads to a generalization of the robustness guarantees derived previously by Lécuyer et al. [2019] and Cohen et al. [2019]. Those certified robustness guarantees specify the radius of an ℓ_2-ball in which the prediction does not change. In contrast, our results specify the robustness that results from the difficulty to identify the ideal attack direction and imply that even for perturbations outside this confidence ball, a gradient-based attack has a chance of not being successful.

To the best of our knowledge, no existing theoretical analysis of SNNs explicitly discusses either the effect of stochasticity during inference nor the impact of the sample size during attack and prediction on the robustness.

3 Preliminaries

We first clarify the terminology before we state the main theoretical results of our paper.

Stochastic classifiers We use the term *stochastic classifiers* for all classifiers which have an inherent stochasticity through the use of random variables in the model. Formally, we define them as follows:

Definition 3.1 (Stochastic classifiers)**.** A stochastic classifier with k classes corresponds to a function $f : \mathbb{R}^d \times \Omega^h \to \mathbb{R}^k$ that maps a pair (x, Θ) to the output $f(x, \Theta) = (f_1(x, \Theta), \dots f_k(x, \Theta))^T$, where $x \in \mathbb{R}^d$ is an input vector, $\Theta \in \Omega^h, \Theta \sim p(\Theta)$ is a random vector, and $f_c(x, \Theta)$ with $c \in \{1, \dots, k\}$ are the discriminant functions for each class. The prediction of a stochastic classifier for an input x is given by $\mathbb{E}_\Theta[f(x, \Theta)]$ and the predicted class by $\arg\max_c \mathbb{E}_\Theta[f_c(x, \Theta)]$.

This generic definition of stochastic classifiers covers linear models with random weights, but also more complicated methods like BNNs [Neal, 1996], infinite mixtures [Däubener and Fischer, 2020], Monte Carlo dropout networks [Gal and Ghahramani, 2016], randomized smoothing as proposed by Lécuyer et al. [2019][1], and any other class of neural networks which use stochasticity at the input level [e.g. Raff et al., 2019] or within the network [e.g. He et al., 2019, Jeddi et al., 2020, Liu et al., 2018, Yu et al., 2021, Eustratiadis et al., 2021]. In cases where $\mathbb{E}_\Theta[f(x, \Theta)]$ is not tractable — which is in practice usually the case — the prediction of the stochastic classifier is approximated by its Monte Carlo (MC) estimate $f^{\mathcal{S}}(x) := \frac{1}{S}\sum_{s=1}^{S} f(x, \theta_s)$, where the samples in the sample set $\mathcal{S} = \{\theta_1, \dots, \theta_S\}$ are drawn from $p(\Theta)$.

Adversarial attacks on stochastic classifiers Informally speaking, adversarial examples are inputs that are modified such that the network predicts wrong classes even though the changes to the inputs are not perceptible for a human. More precisely, let x be an input with corresponding true label $y \in \{1, \dots, k\}$ that is classified correctly by the multi-class classifier $f(\cdot)$, that is $\arg\max_c f_c(x) = y$. We consider the most common attack form which aims at misclassifying x by allowing for some predefined maximum magnitude of perturbation. That is, the attacker targets the optimization problem

$$maximize \quad \mathcal{L}(f(x + \delta), y) \ \ , \ \ s.t. \quad \|\delta\|_p \leq \eta \ \ , \tag{1}$$

where $\mathcal{L}(\cdot, \cdot)$ is the loss function, $\|\cdot\|_p$ with $p \geq 1$ is the ℓ_p-norm, and η is the *perturbation strength*, i.e. the maximal allowed magnitude of the attack (see figure 1a) for an illustration). Common choices of loss functions include the cross-entropy loss and the negative margin loss $\mathcal{L}_{\text{margin}}(f(x + \delta), y) = -(f_y(x + \delta) - \max_{c \neq y} f_c(x + \delta))$. In practice, targeting the optimization problem in eq. (1) usually involves estimating δ by performing some kind of gradient-based optimization on the loss function [e.g. Goodfellow et al., 2015, Madry et al., 2018]. If $f^{\mathcal{S}}(\cdot)$ is a stochastic classifier (as introduced in the previous paragraph) approximated by its MC estimate, the loss gradient with respect to the input is stochastic as well and given by

$$\nabla_x \mathcal{L}(f^{\mathcal{S}}(x), y) = \nabla_x \mathcal{L}\left(\frac{1}{S}\sum_{s=1}^{S} f(x, \theta_s), y\right) \quad . \tag{2}$$

Note, that for linear $\mathcal{L}$, as for example the margin loss for a fixed class c, the loss gradient of the mean prediction is equivalent to the mean of the loss gradients for single sample predictions. However, in general (e.g. for the cross entropy loss) this is not the case.

4 Geometrical robustness analysis

After clarifying our understanding of stochastic classifiers and on how gradient-based adversarial attacks are conducted on them, we are now able to present a simple but general geometrical, adversarial robustness analysis for stochastic classifiers. It is motivated by the following observation: Each time a stochastic classifier is used for a prediction, another set of realizations from the random vectors are

[1] Note, that, in contrast to Lécuyer et al. [2019] the variant of randomized smoothing proposed by Cohen et al. [2019] does not define the prediction of the SNN to be given by the expectation over Θ but by $\arg\max_{c \in \mathcal{Y}} P(f(x + \epsilon) = c)$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. This makes our results not directly applicable to their networks. However, they can probably be transferred to the decision boundaries and attacks corresponding to their form of generating predictions.